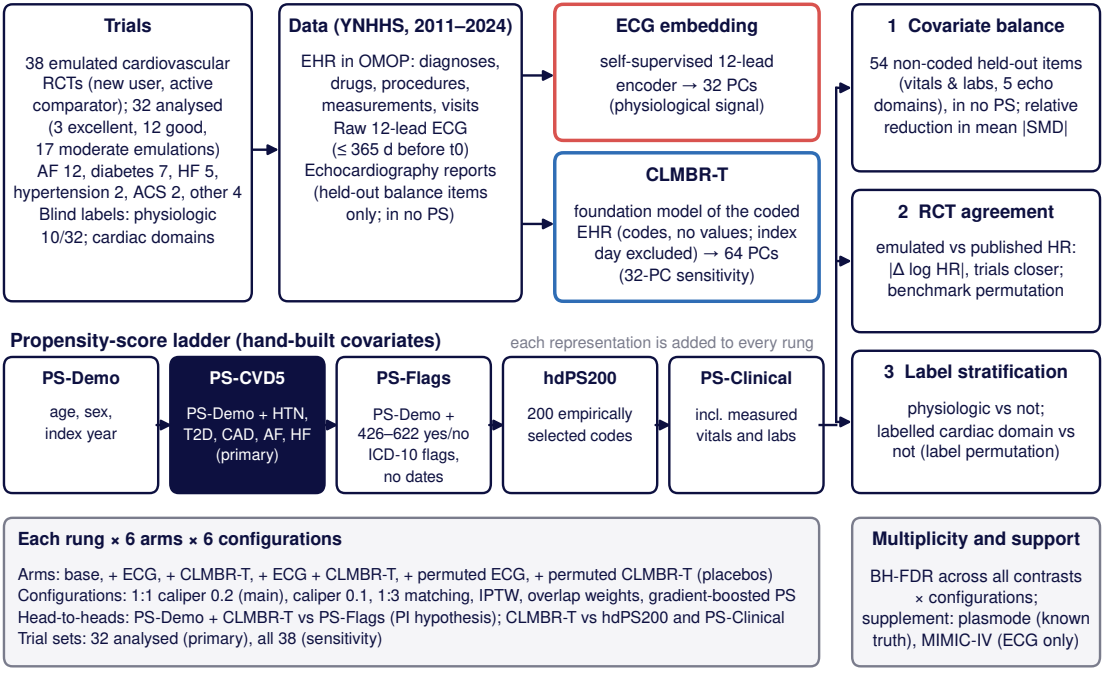
Trials
38 emulated cardiovascular RCTs (new user, active comparator); 32 analysed (3 excellent, 12 good, 17 moderate emulations)
AF 12, diabetes 7, HF 5, hypertension 2, ACS 2, other 4
Blind labels: physiologic 10/32; cardiac domains
Data (YNHHS, 2011–2024)
EHR in OMOP: diagnoses, drugs, procedures, measurements, visits
Raw 12-lead ECG (≤ 365 d before t0)
Echocardiography reports (held-out balance items only; in no PS)
ECG embedding
self-supervised 12-lead encoder → 32 PCs (physiological signal)
CLMBR-T
foundation model of the coded EHR (codes, no values; index day excluded) → 64 PCs (32-PC sensitivity)
1 Covariate balance
54 non-coded held-out items (vitals & labs, 5 echo domains), in no PS; relative reduction in mean |SMD|
2 RCT agreement
emulated vs published HR: |Δ log HR|, trials closer; benchmark permutation
3 Label stratification
physiologic vs not; labelled cardiac domain vs not (label permutation)
Propensity-score ladder (hand-built covariates)
each representation is added to every rung
PS-Demo
age, sex, index year
PS-CVD5
PS-Demo + HTN, T2D, CAD, AF, HF (primary)
PS-Flags
PS-Demo + 426–622 yes/no ICD-10 flags, no dates
hdPS200
200 empirically selected codes
PS-Clinical
incl. measured vitals and labs
Each rung × 6 arms × 6 configurations
Arms: base, + ECG, + CLMBR-T, + ECG + CLMBR-T, + permuted ECG, + permuted CLMBR-T (placebos)
Configurations: 1:1 caliper 0.2 (main), caliper 0.1, 1:3 matching, IPTW, overlap weights, gradient-boosted PS
Head-to-heads: PS-Demo + CLMBR-T vs PS-Flags (PI hypothesis); CLMBR-T vs hdPS200 and PS-Clinical
Trial sets: 32 analysed (primary), all 38 (sensitivity)
Multiplicity and support
BH-FDR across all contrasts × configurations; supplement: plasmode (known truth), MIMIC-IV (ECG only)